

Sampling Techniques in Statistics

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Sampling is the process of selecting a smaller group from a larger **population** to gather information.

Population

A **population** is the complete set of people, objects, or events being studied.

For example, all students in a school can be the population in a study.

Sample

A **sample** is a smaller part of the population chosen for investigation.

For example, 50 students selected from a school can be a sample.

Good Sampling

Good sampling techniques help researchers collect information efficiently while representing the population fairly.

Simple Random Sampling

Simple Random Sampling means every member has an equal chance of being selected. For example, names can be written on slips of paper and randomly drawn from a box.

Systematic Sampling

Systematic Sampling means members are selected at regular intervals after a random starting point.

For example, a researcher may select every 10th person after randomly choosing the first person.

Stratified Sampling

Stratified Sampling means the population is divided into groups called **strata**.

Samples are then taken from each group.

For example, a researcher may select students from every grade level.

Cluster Sampling

Cluster Sampling means the population is divided into clusters.

One or more entire clusters are selected.

For example, a researcher may randomly select two sections and survey every student in those sections.

Convenience Sampling

Convenience Sampling means participants are chosen because they are easy to reach. For example, a student may survey classmates who are nearby.

Examples

Example 1

A teacher writes the names of all 40 students on slips of paper, mixes them in a box, and randomly draws 8 names.

The sampling technique is

Simple Random Sampling

Example 2

A researcher surveys every 12th customer entering a supermarket after randomly choosing the first customer.

The sampling technique is

Systematic Sampling

Example 3

A school has students from Grades 7, 8, 9, and 10. The researcher selects 10 students from each grade level.

The sampling technique is

Stratified Sampling

Example 4

A city is divided into 15 barangays. The researcher randomly chooses 3 barangays and surveys every household in those barangays. The sampling technique is

Cluster Sampling

Example 5

A student surveys the first 20 classmates she meets during recess.

The sampling technique is

Convenience Sampling

Practice Exercises

Practice 1

A teacher writes the names of all 40 students on slips of paper, mixes them in a box, and randomly draws 8 names.

What sampling technique is used?

Practice 2

A researcher wants to survey every 12th customer entering a supermarket after randomly choosing the first customer.

What sampling technique is used?

Practice 3

A school has students from Grades 7, 8, 9, and 10. The researcher selects 10 students from each grade level.

What sampling technique is used?

Practice 4

A city is divided into 15 barangays. The researcher randomly chooses 3 barangays and surveys every household in those barangays. What sampling technique is used?

Practice 5

A student surveys the first 20 classmates she meets during recess.

What sampling technique is used?

Practice 6

Identify whether the following statement is about a **population** or a **sample**.

“All Grade 7 students in the school.”

Practice 7

Identify whether the following statement is about a **population** or a **sample**.

“Thirty Grade 7 students selected from the school.”

Practice 8

Which sampling technique gives every member of the population an equal chance of being selected?

Practice 9

Why is stratified sampling useful when the population consists of different groups?

Practice 10

Match each situation with the correct sampling technique.

(a) Surveying every 8th visitor entering a museum.

(b) Drawing names from a box.

(c) Selecting all students from two randomly chosen sections.

(d) Surveying only nearby shoppers.

(e) Choosing equal numbers of boys and girls.

Activities

Activity 1

A coach writes the names of all 30 players on cards, shuffles them, and randomly selects 6 players.

What sampling technique is used?

Activity 2

A researcher surveys every 15th passenger entering a bus terminal after choosing a random starting passenger.

What sampling technique is used?

Activity 3

A university has students from four colleges. The researcher selects 20 students from each college.

What sampling technique is used?

Activity 4

A province is divided into several municipalities. The researcher randomly selects 4 municipalities and surveys all households in them. What sampling technique is used?

Activity 5

A vendor asks the first 25 customers who visit the store to answer a questionnaire.

What sampling technique is used?

Activity 6

Identify whether the following statement is about a **population** or a **sample**.

“All teachers in the district.”

Activity 7

Identify whether the following statement is about a **population** or a **sample**.

“Forty teachers selected from the district.”

Activity 8

Which sampling technique involves selecting participants at regular intervals from a list?

Activity 9

Why is cluster sampling useful when the population is spread over a large geographic area?

Activity 10

Match each situation with the correct sampling technique.

- (a) Selecting every 5th patient arriving at a clinic.
- (b) Randomly drawing employee ID numbers.
- (c) Surveying all families in two selected villages.
- (d) Asking only friends to answer a survey.
- (e) Selecting equal numbers of students from each grade level.

Thank you!